Building a Training Set for Drone-and-Buggy Litter Detection

Gap analysis of the current image set and a capture plan

August 5, 2026

Summary

The target system is a three-stage pipeline: a 4K drone scans from high altitude, descends for a low-altitude pass, and a ground buggy with a camera and arm retrieves the objects. Each stage sees objects at a completely different scale, from a completely different angle, and the current image set supports *none of them*.

Three findings drive everything below.

1. **There is no aerial imagery at all.** All 244 images are handheld iPhone photographs taken from roughly waist height. The directory scaffold for aerial, occlusion, lighting and weather variation exists but contains only 27 `.gitkeep` placeholder files and zero images.

2. **Detecting cigarette butts from 20 m is not possible with a 4K camera.** This is a physical limit, not a model-quality problem (Section 2). At 20 m a butt spans about 3 pixels. No amount of training fixes that. The mission profile has to be restructured around what each altitude can actually resolve.

3. **The set is too small and too unbalanced to train on.** After classification, 107 of 244 images contain no identifiable object, and 13 of the 23 classes have five examples or fewer. Useful object detection needs on the order of 10^3 instances per class, not 10^0.

The good news is that the hard part — knowing what the target objects look like in situ, and a working manual polygon annotation workflow — already exists and should be preserved.

1 What the dataset currently is

Property	Value
Images	244
Capture device	iPhone SE (3rd gen), 242/244
Resolution	4032 × 3024 (12 MP), 240/244
Images with GPS EXIF	242
Viewpoint	Handheld, ground level, near-nadir
Altitudes represented	One (≈1.2 m handheld)
Manually annotated files	100
Manual polygons	331
— of which are background (rock, grass, road, sky)	205
— of which are waste objects	126
Classes after auto-classification	23 (+ dontknow)

Table 1: Current state. Every image comes from a single camera at a single height, which means the model cannot learn scale or viewpoint invariance from this data.

Class distribution is severely long-tailed:

Class	n	Class	n
dontknow (no object found)	107	glove	4
cigarette_butt	39	candy_wrapper	4
plastic_bottle	17	can	4
plastic_bag	14	pine_cone	3
plastic_wrapper	7	chair	3
textile, plastic_piece, paper	6 each	bottle_cap	3
straw, container, cig_pack	5 each	6 further classes	1 each

2 The resolution limit sets the mission profile

Ground sample distance (GSD) is how much real-world length one pixel covers:

$$\mathrm{GSD} = \frac{2h\tan(\theta/2)}{W}$$

for altitude h, horizontal field of view θ, and image width W pixels. For a 4K sensor ($W = 3840$) with a typical 24 mm-equivalent drone lens ($\theta \approx 73°$):

Alt.	Swath	GSD	Object size in pixels (longest dimension)				
(m)	(m)	(mm/px)	Cig. butt	Cap	Wrapper	Bottle	Chair
20	29.6	7.71	**3.2**	**3.9**	**7.8**	32	104
10	14.8	3.85	**6.5**	**7.8**	**15.6**	65	208
5	7.4	1.93	**13.0**	**15.6**	31	130	415
2	3.0	0.77	32	39	78	324	1038
1	1.5	0.39	65	78	156	649	2076
0.2	0.3	0.08	324	389	778	3243	10379

Table 2: Red entries fall below roughly 20 px, the practical floor at which small-object detectors still work. A cigarette butt is about 8 × 25 mm.

Inverting this gives the **maximum altitude at which each object class is still detectable** (20 px threshold):

Object	Max. altitude
Cigarette butt	3.2 m
Bottle cap	3.9 m
Wrapper	7.8 m
Plastic bottle	32 m
Chair / large debris	100 m+

Consequence for the planned mission

The proposed profile — scan at 20 m, then 1–2 m, then buggy — needs one correction: **the 20 m pass cannot be a litter-detection pass**. It should be re-scoped as a *survey* pass that produces:

- detections of large items only (bottles, bags, cans, furniture);
- a terrain/surface map (grass vs. gravel vs. asphalt) used to prioritise;
- a coverage plan for the low pass.

Small-object detection has to happen on the **1–2 m pass**, where a butt is 32–65 px. Note the cost: at 2 m the swath is only 3 m wide, so covering one hectare takes roughly 3.3 km of flight track. Budget the low pass over prioritised regions, not the whole area — which is exactly what the high pass should be producing.

A useful lever: if the drone can carry a longer lens or crop-zoom, the 20 m pass gains proportional resolution. Doubling focal length halves GSD and halves the required low-pass coverage. Worth testing before committing to a flight plan.

3 What needs to be captured

Targets below assume a detector trained per class. Quantities are *annotated object instances*, not images; one image can carry many.

3.1 Altitude and viewpoint coverage

Tier	Height	Angle	Purpose
Buggy-low	0.2 m	0–30° oblique	Final approach, arm alignment
Buggy-high	1 m	nadir + 45°	Buggy search/navigation
Drone-low	2 m	nadir	Primary small-object detection
Drone-low	5 m	nadir + 30°	Transition scale
Drone-high	10 m	nadir	Large-object detection
Drone-high	20 m+	nadir	Survey, terrain prioritisation

The single most valuable capture method is the *altitude ladder*: place a known object, then photograph the identical scene at 0.2, 1, 2, 5, 10 and 20 m without moving anything. This yields exact cross-scale correspondence, lets you measure empirically where each class stops being detectable, and supports scale-augmentation training. Repeat per surface type. Roughly 50 ladders would transform the dataset's usefulness.

Angles matter because the buggy sees objects obliquely while the drone sees them from directly above — these look nothing alike. Every class needs both.

3.2 Occlusion

Currently unrepresented as a labelled axis. Capture each class at four levels, and **record the occlusion fraction in the annotation** so it can be evaluated separately:

Level	Definition
None	Fully visible against surface
Light	$< 25\%$ hidden; grass blades crossing the object
Partial	25–60% hidden; in grass tufts, under leaf edges
Heavy	$> 60\%$ hidden; only a fragment protrudes

Occluders to cover explicitly: tall grass, cut grass, dead leaves, loose soil, gravel, twigs/pine needles, puddles, and *other litter* (overlapping objects, which the pipeline will meet in dumping hot-spots).

3.3 Surface, lighting, weather

Backgrounds already present are gravel, soil and grass. Missing and needed: asphalt, concrete/paving, sand, wood chip, leaf litter, and short mown lawn. Surface matters more than usual here because low-contrast pairings are the hard cases — a beige butt on gravel, a clear bottle on wet asphalt.

Lighting in the current set is 230/244 bright sunlight. Required additions: overcast, deep shadow, dappled shade under trees, golden hour with long shadows, and dawn/dusk. Add wet-surface and post-rain conditions. If night operation is intended, capture under drone illumination.

3.4 Negatives

The 107 `dontknow` images are an asset, not waste — they are **hard negatives**: real ground with no litter. Keep them as explicitly labelled empty frames. Add deliberate confusers: pine cones, stones, leaves, bark, dog waste, twigs, mushrooms, and flowers, all of which resemble litter at low pixel counts. Target 20–30% of the final set as negatives.

3.5 Volume targets

Class tier	Instances (minimum)	Instances (comfortable)
Priority (cigarette butt, bottle, can, wrapper, bag)	1,000	3,000
Secondary (cup, cap, straw, glove, textile, paper)	500	1,500
Rare / large (chair, crate, furniture)	200	500
Hard negatives (empty ground, confusers)	2,000	5,000

Each priority-class instance count should be spread across the altitude tiers, not concentrated at one height. A practical rule: no more than 40% of a class's instances from any single altitude tier.

4 Annotation requirements

The existing polygon workflow is sound and should continue — polygons are worth the extra effort here because they support segmentation, and grasp planning for the arm needs object outline, not just a box.

Changes needed:

1. **Fix the label vocabulary.** Current labels mix Danish and English (`serviet`, `krukker`, `grankogler`) and contain typos and truncations (`dirty_clot`, `dirt_r`, `bottle_`, `gra`, `1`, `2`). Freeze a controlled vocabulary and validate against it at annotation time.
2. **Separate background from object classes.** 205 of 331 polygons label terrain (rock, grass, road, sky). These are useful for a segmentation task but must not be mixed into the detector's object classes.
3. **Record per-instance metadata**: altitude, camera angle, occlusion level, surface, lighting. Without these the evaluation in Section 5 cannot be broken down, and you will not know *where* the model fails.
4. **Add a scale reference** to capture sessions — a coloured marker of known size in frame lets you verify GSD and recover real object dimensions, which the arm needs for grasp sizing.
5. **Verify a sample by hand.** Spot-checks during this analysis found manual labels that disagree with the imagery (one `paper` polygon contains a drinking straw; `dirty_cloth` is most likely automotive carpet). Budget a second-pass review.

5 Test videos for offline evaluation

Video is the right evaluation medium because the deployed system sees motion, motion blur and repeated views of the same object — and because a video lets you measure *tracking consistency*, which single images cannot.

General recording rules. Record 4K at 30 fps, in the highest-quality codec available, with stabilisation *off* for at least half the takes (the model must tolerate real vibration). Log altitude, speed and GPS. Before each take, place a known object set and photograph a ground-truth layout map, so every object in the scene has a known identity and position. Keep takes to 60–90 s.

Required takes

Take	Profile	What it measures
Altitude descent	Hover, descend 20 m→1 m over a fixed seeded plot	The empirical detection threshold per class. The altitude at which each object first appears is the single most useful number you can collect.
High survey lawn	20 m, 3 m/s, straight transects	Large-object recall and false-positive rate over open ground.
Low sweep	2 m, 1 m/s, parallel tracks with 20% overlap	Primary small-object detection performance; also validates that the swath/overlap plan actually achieves full coverage.
Speed sweep	2 m at 0.5, 1, 2, 4 m/s	Where motion blur breaks detection — sets the operational speed limit.
Occlusion plot	2 m over a plot seeded at all four occlusion levels	Recall as a function of occlusion; the number to quote when scoping expectations.
Surface transition	2 m crossing grass→gravel→asphalt	Background-shift robustness within a single continuous clip.

Take	Profile	What it measures
Lighting series	Same plot at dawn, noon, overcast, dusk, wet	Illumination sensitivity with all else held constant.
Buggy approach	1 m→0.2 m driving toward a target	The handoff: does the object stay tracked as it grows and the angle flattens?
Buggy patrol	0.5 m, continuous drive through a seeded area	End-to-end ground-stage recall and duplicate-detection behaviour.
Grasp approach	0.2 m static, arm entering frame	Whether detection survives occlusion *by the robot's own arm* — a failure mode that is easy to forget and fatal in practice.
Confuser plot	2 m and 0.5 m over pine cones, stones, leaves only	False-positive rate on a scene containing zero litter.
Empty control	Each altitude over verified-clean ground	Baseline false-positive rate. Any detection here is an error, which makes this the cheapest take to score.

Seeded plots. Lay out a 5×5 m plot with a known inventory — e.g. 20 cigarette butts, 5 bottles, 5 cans, 10 wrappers, 3 bags — at recorded positions. Because the count is known exactly, recall can be computed without annotating every frame. Build three plots: sparse (realistic), dense (hot-spot), and confuser-heavy.

Scoring the videos

Annotate every N-th frame (10–30) rather than all frames, and report precision/recall broken down by altitude, occlusion level, surface and class — aggregate numbers will hide exactly the failures that matter. Track two video-specific metrics: **re-detection stability** (fraction of frames in which an object, once seen, stays detected) and **time-to-first-detection** during descent.

Keep every test video strictly out of training. Given GPS is recorded, split train/test **by location**, not randomly — random splits across frames of the same site inflate scores badly.

6 Suggested order of work

1. Freeze the label vocabulary; fix the existing typo/language labels.
2. Fly one altitude-descent test video. It costs an afternoon and tells you the real per-class detection ceiling before you invest in bulk capture.
3. Decide the mission profile from that measurement (Section 2 gives the prediction; the flight gives the truth).
4. Bulk-capture the altitude ladders across surfaces and lighting.
5. Annotate with the extended per-instance metadata.
6. Record the full test-video suite; keep it sealed from training.

Figures in Sections 1–2 are computed from the current repository contents and from the stated 4K/24 mm-equivalent camera assumption; substitute the real lens parameters when the airframe is chosen, as GSD scales linearly with focal length.